

World's Smallest Programmable INDUSPHONE DESIGN

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In the previous article of this DIY device series, the smallest touchscreen phone was created using the SIM800L for the 2G cellular network and a round touch

display. This design is now finalised by integrating the previously developed user interface (UI) with its core functions and consolidating everything into a single system powered by a real-time operating system (RTOS). The cellular network compatibility is also enhanced by

replacing the 2G GSM module with a 4G GSM module, enabling support for LTE and 4G networks (see Fig. 2). Additional applications, such as a calculator and other basic apps, are included. The entire system operates on the RTOS platform. Fig. 1 illustrates the IndusPhone



Fig. 1: IndusPhone calculator application

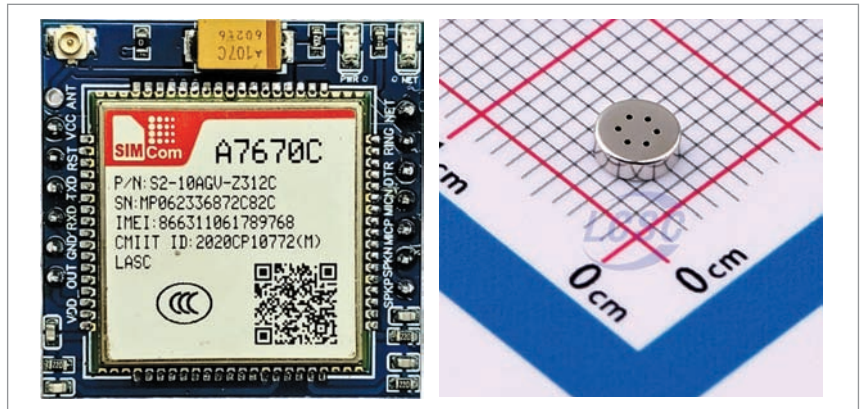


Fig. 2: 4G GSM module (left) and MIC (right)

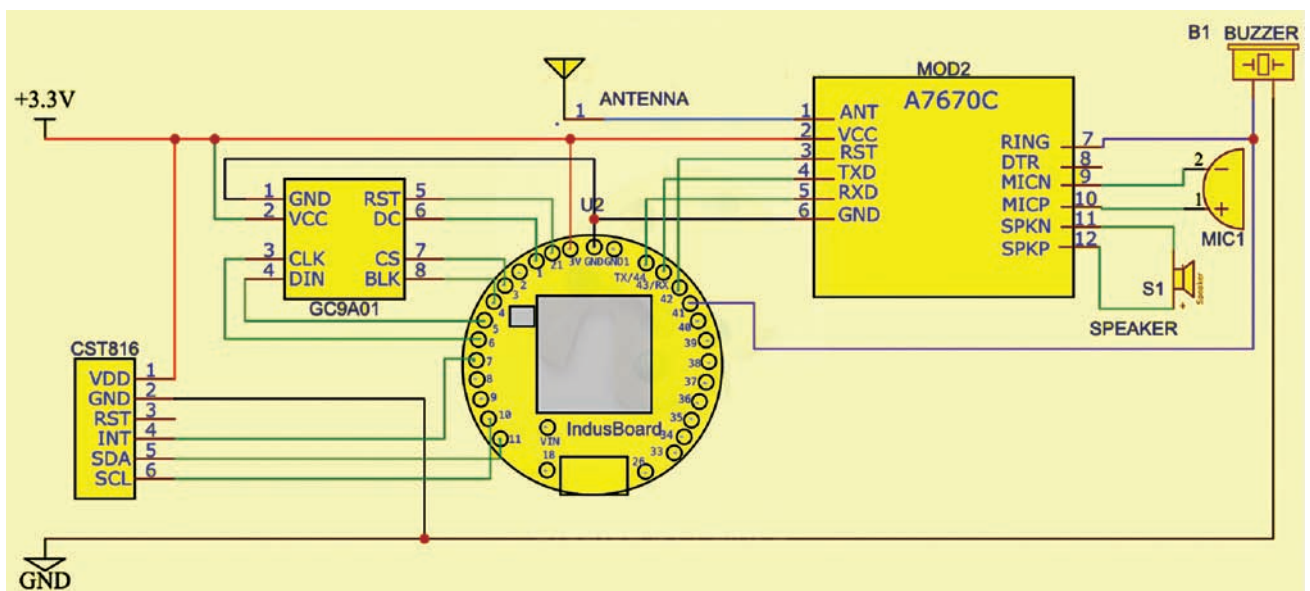


Fig. 3: Circuit diagram